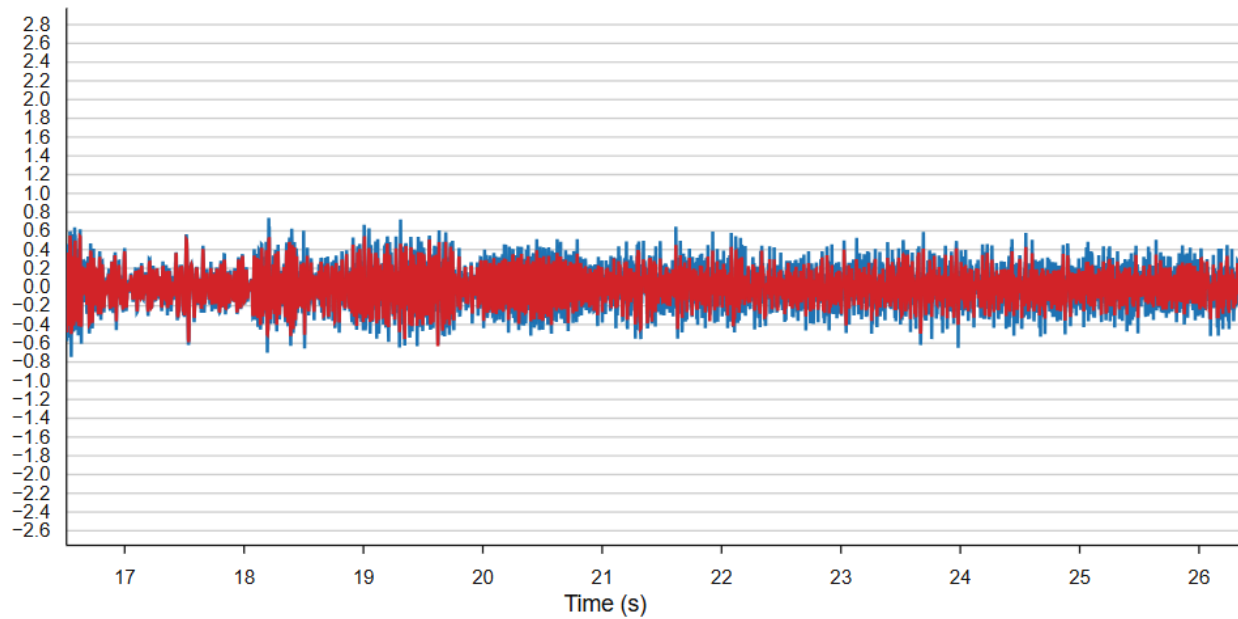


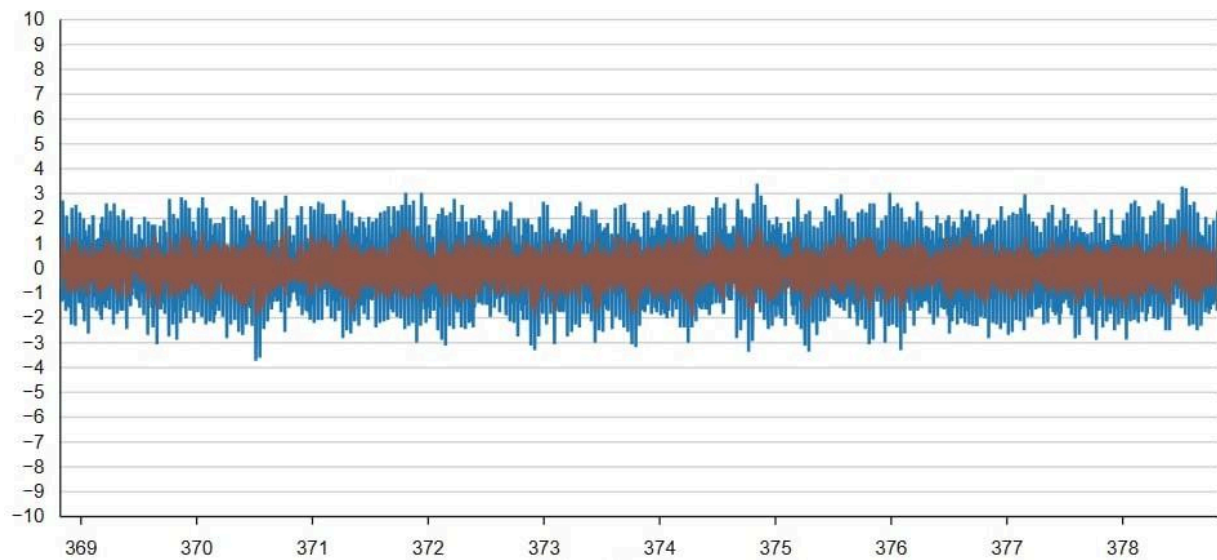
Filter Result

1) Notch Filter and LowPass Filter



Where as:

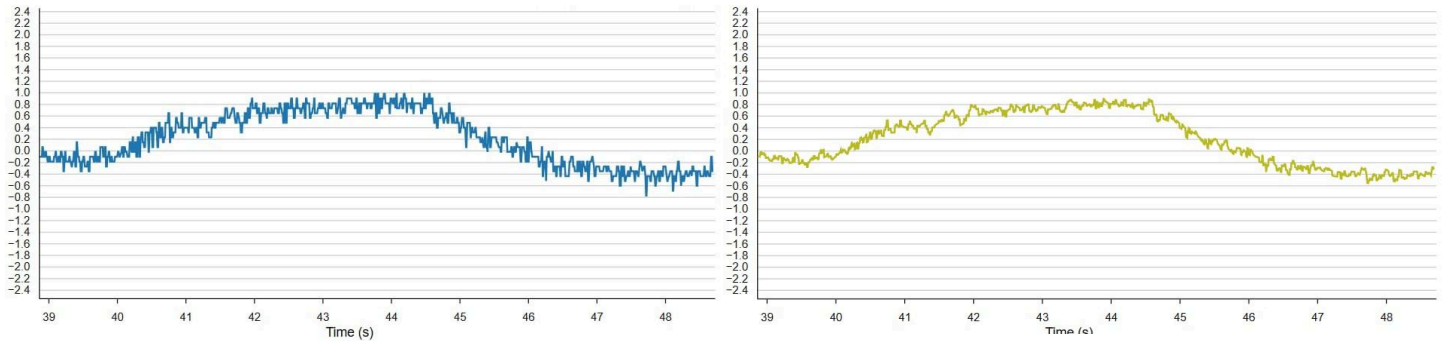
- Accelerometer data calculated from IMU
- Lowpass Filter accelerometer data



Where as:

- Gyroscope data calculated from IMU
- Notch + Lowpass Filter gyroscope data

2) Kalman Filter



Where as:

— Height value calculated from barometer

— Height value from Kalman Filter

Flight Result

Video:

<https://youtu.be/znj9OqjzdcU>

The quadcopter has a stable flight between 0.1m - 4m height from the ground.

The PID control enables reaction immediately.

Pitch, roll and yaw maintain between +5 and -5 value from the setpoint.

It still can't hold in one position as the PID parameter is not the most perfect and has no optical flow sensor or GPS.

It can maintain the desired height based on the Kalman Filter height estimation derived from barometer data.